

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-2	Team No-24	Project Title- The Vehicle Service Booking and Tracking System
----------	------------	--

1. Abstract

The Vehicle Service Booking and Tracking System is a web-based application designed to digitize and streamline vehicle servicing at automobile service centers. Traditionally, customers rely on phone calls or walk-ins to book appointments, leading to scheduling conflicts, long wait times, and little visibility into their vehicle's status during servicing. This project addresses these issues by providing a centralized platform where customers can register vehicles, book service slots online, select required services, and track real-time status, from check-in and inspection through repair, quality check, and delivery. Service center staff and administrators get a dashboard to manage bookings, assign mechanics, update job status, and maintain each vehicle's service history. The system uses a relational and distributed database backend to ensure data consistency, reliability, and efficient handling of concurrent bookings across multiple service centers. Key features include automated slot allocation, SMS and email notifications, digital service history records, invoice generation, and role-based access for customers, staff, and administrators. By replacing manual, paper-based processes with an automated, transparent, and scalable system, the platform aims to improve customer satisfaction, cut administrative overhead, and let service centers operate more efficiently.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030424	Revanth Reddy	
2520030391	Subhash	

3. Frameworks / Platforms / Software Used

- **Frontend:** React.js (HTML, CSS, JavaScript)
- **Backend:** Node.js with Express.js
- **Database:** MySQL / PostgreSQL, with Redis for caching
- **Authentication:** JWT / OAuth2
- **Notifications:** Twilio API (SMS), Nodemailer (Email)
- **Version Control & Deployment:** GitHub

Guide Name and Signature (remarks if any) _____